

Richard Orilla

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Education

De La Salle University	2017–2019
M.Sc. in Computer Science (Coursework Completed)	
University of Mindanao	2010–2015
B.Sc. in Computer Science (Status: Graduated)	

Experience

inRiver	Present
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Technical Lead

- Architected 5 interconnected microservices (CMS, Central Command, Spider, ETL, Post-Processing) using Python and Azure DevOps CI/CD to manage enterprise-scale web scraping workflows with isolated, per-customer databases.
- Led a team of 8 engineers in designing the complete data pipeline and designed SQL Server schemas for customer data synchronization via CMS with real-time replica databases, ensuring zero cross-customer data leakage.
- Evolved the architecture into an AI Agent Orchestrator using Python, and engineered an interactive client-side AI Chat UI using React that manages session state purely in-browser to comply with strict EU data privacy laws.
- Spearheaded the design of a C# data mapper to interface the core C# PIM platform with Python services, facilitating seamless post-acquisition corporate technology integration.
- Coded Vue + Quasar front-ends and managed the transition to modern React applications designed with Material UI (MUI).
- Standardized containerized infrastructure using Docker and Kubernetes (Helm) to reduce deployment friction across staging and production environments.

Detail Online	2018–2022
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Senior Backend Developer

- Maintained high-performing NodeJS and JavaScript backend applications to process compliance results, building custom query engines that enabled efficient dynamic client-side sorting and row-fetching.
- Engineered a portable C#/.NET Windows desktop application to automate and streamline data validation QA workflows for internal teams.
- Built ML-powered automation tools in Python using TensorFlow, Caffe, and Darknet to eliminate manual data processing, saving 8 hours of labor overhead per week.
- Designed and delivered 15+ production REST APIs in Python interacting with PostgreSQL databases to service organization-wide data pipeline integrations.
- Conducted technical evaluations of third-party AI platforms and prototyped vision-based ML models in PyTorch to benchmark capabilities against requirements, saving an estimated \$50K in licensing costs.

University of Mindanao	2015–2018
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Computer Science Instructor

- Instructed academic coursework covering Fundamentals of Programming, Object-Oriented Programming (OOP), Database Programming (SQL Server/IBM DB2 fundamentals), Visual Programming, Game Development, and Security.

- Developed curricula, prepared lectures, and mentored over 100+ undergraduate students in software engineering and database design concepts.

Freelance Software Developer

2015–2018

Independent Contractor

- Maintained legacy business-critical C++ desktop applications integrated with MySQL, resolving bugs and implementing minor feature updates (2016–2017).
- Developed custom WordPress PHP plugins and maintained Laravel e-commerce web applications to support client online scaling and feature updates.

Technical Skills

Languages: Python, C#, JavaScript, SQL, C++, PHP, HTML, CSS

Frameworks & Libraries: React, Vue, NodeJS, Quasar, Material UI (MUI), Laravel, TensorFlow, PyTorch

Tools & Database: Docker, Kubernetes, Helm, Git, Azure DevOps, SQL Server, PostgreSQL, MySQL, IBM DB2

Projects & Research

DOSBox Staging Replacer

C++

Open-source utility that replaces GOG Galaxy's default DOSBox emulator, resolving scaling and performance issues for retro gaming on modern platforms.

Amagon HTML Editor

Electron, Monaco Editor, AI

Offline visual editor featuring bidirectional code/canvas sync and natural language page building.

Team Report Generator

Laravel, Gemini AI

LLM-driven daily update summarizer with secure JSON proxy, CORS, and rate limiting.

s2sc.js Transpiler

Compiler Construction

Source-to-source compiler exploring language interoperability via AST parsing and code generation.

Hand Tracking Game

Computer Vision, 3D Physics

Browser-based game using MediaPipe landmark detection mapped to React Three Fiber + Rapier.